

Development Prompts — NL2SPICE Project

This file lists all natural-language prompts entered during the development of the NL2SPICE pipeline in this project directory. Prompts are grouped chronologically by development phase.

Phase 1 — Baseline Pipeline Setup

1. Can you see `test_load.py` and `chat_demo.py` in `Course_Projects/CPSC5770/LLM_Trial`? That basically shows that we have an LLM loaded onto our cluster and we can invoke it using Python. In `Course_Projects/CPSC5770/Project_Baseline/Test_Data` we have a bunch of JSON files. These JSON files have prompts and [specs]. I want you to build a pipeline that feeds each prompt to the model and evaluates the output.
 2. Does this script try to parallelise runs? I will try to get a job with more than 1 GPU.
 3. Yes please, I will try to get 4 GPUs. What is the `batch_size` number you recommend for however many GPUs I get?
 4. And what performance metrics are you tracking?
 5. And are these metrics being logged in the output `.json` files?
 6. How large might the files get?
 7. Okay not bad. Also since I will try to submit this through a Slurm job, is there any way I can monitor this by just looking at some files on the login node? Can you make something like that?
 8. Okay great. One final question, how long might this take? Because I can only get GPU jobs for 6 hours.
 9. Okay I will test out a small job and let you know.
-

Phase 2 — Running & Debugging the Baseline

10. (*Pasted pipeline progress output*) Look at the outputs now.
 11. Look at it now.
 12. What about now?
 13. Check now?
 14. Check now? Also why is the `fc_error` so large?
 15. Are we sure that the model is actually that bad? Is there a way to verify this? I know it's better for us that the model is, but I hope it's not this bad always.
 16. Okay now I can run the full job right, or shall we run a larger test?
 17. Check now and tell me if I can go ahead for the full job.
 18. Check now with the fixes. And I can get a 24-hr job. This is with 2 GPUs, I can get 4, so how long might that take?
 19. Why is `progress.txt` not being updated with the latest run?
 20. Now if I close this chat, will it be saved?
-

Phase 3 — Results Analysis & Plotting

21. Okay can you go through the outputs and write a script to neatly analyse and plot the metrics? Can you see the outputs?
 22. Why is only LPF specified in the cutoff frequency graph?
-

Phase 4 — Agentic Loop Implementation

23. Look at this proposal, now I basically want to implement the agentic loop part of the project. Where the LLM can call NGSpice and run simulations and end up giving a better output since these outputs [are fed back]. (*Attached project proposal.*)
 24. How are these feedback messages generated? Also keep in mind we will have a formula sheet of sorts in the final pipeline when the RAG is implemented as well. So for now, try not to give exact hints or formulas.
 25. Okay good, give me a command to run this on a GPU cluster. Also where are these outputs stored? Make a new output folder if you can.
 26. All the specs seem to be missing, is that expected? Also are you logging all the iterations somewhere? If not please log it so that I can go through it. Please make a new output folder called `agentic_output`.
 27. When I said specs are missing I meant the model is not meeting the specs.
 28. `"ngspice_stderr": "Note: v1: has no value, DC 0 assumed\n"` — Is this error okay?
 29. The performance is still quite bad, will the RAG really be able to help us?
 30. Okay add a chain-of-thought prompt to the current `agentic_pipeline` and let's see how well it performs.
 31. Check the outputs now.
 32. What about other topologies, do they work appropriately as well?
 33. Also does this mean our baseline metrics from before were incorrect as the SPICE simulations might be wrong?
 34. Did you re-run the code to get these metrics?
-

Phase 5 — New Data Generation

35. If you look at the new folder structure in `Course_Projects/CPSC5770`, you will see a new folder called `New_Datagen`. I want you to go through that and see if you understand it. Once you do that, I want [you to generate datasets for all four filter types].
36. You should be able to find the NGSpice path `NGSPICE_PATH = "ngspice"`, do `module load miniconda && conda activate cp5770_project` and it should be available.
37. How long do you think it will take?
38. Okay in the meantime, do you remember the baseline we did?
39. Wait, where is the BPF dataset stored?
40. Okay that is fine, I got confused. Can you now make a new `pipeline.py` that is similar to `Course_Projects/CPSC5770/Archive/pipeline.py` that will test the prompts we just generated?
41. Okay I ran it, can you see the outputs?
42. Okay I have another issue, can you modify the system prompt to add a line of requirements so that later in the agentic loop, the agent can look at said requirements to see whether it is meeting them or not.
43. Hmm okay so I have a doubt — when the agent is reiterating, is it looking at the requirements from the generated answer or the dataset JSON?

44. Yes, add it to the user prompts for the dataset generation for now, generate the 4000 prompts again and modify the pipeline please, let's look at the agent later.
 45. Is it done now?
 46. So pipeline need not be changed right? Because the evaluation will only extract the netlist from the answer.
 47. And where did you add these "add requirements block" in the prompts?
 48. Is the generation done now?
 49. Where are the outputs of pipeline_v2 stored?
 50. Okay I ran a small test, can you verify whether the requirements are being generated?
 51. I ran the pipeline_v2 code, do you see the new outputs?
 52. What does "functional" mean?
 53. Okay now do you remember the performance metrics we calculated for the old baseline?
 54. No I meant this file `Course_Projects/CPSC5770/Archive/analyse_results.py`. Can we improve on the analysis now that we actually have the ground truth netlist as well?
 55. Can you see the PPT here `Course_Projects/CPSC5770/New_Datagen/output/analysis/`?
 56. I've uploaded a PDF version of the PPT here, I want you to go through slides 18–22 and tell me what to edit please.
 57. Yes please and also elaborate a bit on the datagen we did. Also I think you need to explain performance metrics on Page 20. What other changes are required?
-

Phase 6 — Presentation Figures

58. Give me a simple Bode plot for this.
 59. Just give me a simple Bode plot for any response, I just need something to put in my presentation.
 60. In Fig 8 the density plot shows that quite a few netlists lie near the ideal line but it's functionally wrong, why is that?
 61. Is that the best graph we can use, or is there another one?
 62. Let's go with A.
 63. Can you make the plot look more professional?
 64. No, keep the old format of only the scatter, just make the font look better and the points pop a bit more.
 65. What is a good background I can keep for these slides? Something like a circuit which will look good in the background.
 66. What progress can I write for the agentic simulations?
 67. Okay check the presentation PDF now and see if everything is okay. Don't worry about slide 21, I have added animations. Also give me a script for each slide please. I want to cover these slides in 4 minutes.
-

Phase 7 — RAG Pipeline (Static Lookup)

68. Now look at this folder `Course_Projects/CPSC5770/RAG` stuff. Let us use these documents to provide context to the agent assuming it is coming from a RAG and then look at the outputs. Do you understand [how to wire this in]?
69. So does it only load the relevant doc for that circuit or everything?
70. Okay so I can run it for the entire dataset and analyse the results later like pipeline_v2?

71. I just ran a small test, can you see the outputs and see how long it took and then verify how long it might take for the entire dataset?
 72. How is the prompting being done? You are using the requirements block to understand how far we are from the required values right? Not from the input JSONs.
 73. But the agent won't have the JSON file in the final testing right? So we need to look at the requirements block.
 74. Yes please let's go with B.
 75. Okay I just ran another test, why do you think bandpass is failing?
 76. Okay that is fine, we can analyse this later after a full run right?
 77. Okay give me a command to run 600 points.
 78. Is it doing the multi topologies as well?
 79. *(Pasted LPF run output.)* — I have a large GPU, can we make this faster?
 80. If I get 2 GPUs, how long will that take?
 81. `pip install flash-attn --no-build-isolation` — So when I run this should I `module load cuda` as well?
 82. `module load CUDA/12.1.1` — *(Following up on the above.)*
 83. Okay I will submit an sbatch with 2 GPUs, give me the commands to test across 1000 points.
 84. *(Pasted GPU output / job script.)* — *(Checking status.)*
 85. *(Pasted cat agent_pipeline.txt output.)* — The memory is fine, but can you make the stdout better?
 86. Okay good, thank you!
 87. Okay I ran several tests, I think notch ran very slow so I could not do the entire dataset, can you go through the outputs and compile them for analysis!
 88. Okay write a Python script to analyse — and why do you think bandpass is failing so bad?
 89. Why is the BPF failing, could we increase context size to make it better or is the .md file not great?
 90. Okay also the input prompt JSONs also have a ground truth netlist, is there a way to compare the generated netlists with that? Is there a metric to measure similarity?
 91. Do you need more CPU cores for NGSpice?
 92. Why do you need to generate again if we have the netlists already?
 93. Yes please, let's implement all three metrics.
 94. Can you give me the command to run? Also what JSONs are you verifying?
 95. Please rename the JSONs we want to analyse to something readable so that I can reference it. And where are you getting the ground truth netlist? Also give me the Python command to run.
 96. Can you also add a check to see whether the ground truth netlists meet the spec?
 97. *(Pasted Freq Pearson r: nan output.)*
 98. Continue from where you left off.
 99. Please try again.
-

Phase 8 — Metric Revalidation & RAG Re-evaluation

100. *(Pasted analysis-by-task output.)* — *(Reviewing results.)*
101. Okay good, now it's better: *(pasted updated metrics table.)* — We should look into Notch GT match as well.

102. Okay now should we redo the rag_pipeline test? Because the model might be correct. And we just need to reanalyse the pipeline outputs right?
103. *(Pasted old vs new spec % comparison.)* — Yes please.
104. Okay how do we fix this? I suppose we need to modify the .md files to reflect the prompts, because the requirements block matches the prompts right?
-

Phase 9 — RAG Fine-Tuned Final Pipeline

105. Okay so is the rag_pipeline fixed, I can keep another run?
106. I just ran another rag_pipeline yesterday, can you see the results?
107. Let us hold on .md files for now. Instead let us reevaluate how we are checking our metrics. Does the method shown in the .md files, the prompt generation files, and our pipeline match?
108. Can we revert the .md files back to what they were? Because I will get a RAG that is fine-tuned on those documents. Or should I copy paste them myself?
109. Okay so now I've replaced it with the original versions, everything matches right?
110. So now we should get better results on the RAG right?
111. I meant agent.
112. Okay give me a command to run a small test of 100 points in total, across all combinations equally.
113. Also should we re-run the baseline?
114. *(Pasted run output.)* — So the feedback is sort of overriding the .md file, is it?
115. Hmm okay, let us modify the .md files again and test it, let's see how well it performs.
116. Had the feedback been fixed as well?
117. Why is the basic LPF and HPF not working? That should be the easiest to work right?
118. *(Request interrupted / re-submitted.)* — Sorry please continue, I clicked No by mistake.
119. Continue from where you left off.
120. I ran a few more tests. Please do look at it.
121. Okay do fix that. And also now look at the RAG_final folder, it has the RAG model that we will use for the final pipeline. Give me the command to run the agentic pipeline to see the feedback changes and RAG retrieval happening.
122. Where is the final pipeline file? Have you created it?
123. Have you gone through the RAG_final folder? Do you know how to use the RAG model in that folder?
124. Can you change the directory of the .md files used by the RAG model and use our modified files. And let us test if the model retrieves the correct files from some prompts.
125. Is the code retrieving the document every iteration?
126. *(Pasted LPF run output.)* — *(Reviewing retrieval behavior.)*
-

Phase 10 — Diagnosis & Notch/BPF Fixes

127. Try to look at this issue with fresh eyes. Why is it that even with proper documents and feedback, the model is achieving only about 50% spec pass? Isn't this an easy problem to solve for an agent, especially with an explicit feedback loop?
128. Yes let's do that!

129. Does the feedback netlist have a fixed template or does it calculate it for that case?
130. And does it take that spec from the requirements block? Also does this eliminate the need for the model at all?
131. So the netlist is being generated by the model and then the feedback changes it?
132. What do the pb and sb numbers mean?
133. *(Shared results_lpf_20260429_164044.json output.)* — This is the output for the rag_final_pipeline, can you check this?
134. But are these results true or is it being hallucinated? Can we verify it's real?
135. Did you re-run the NGSpice sims to see that?
136. Okay that's great!
137. *(Shared HPF results.)* — Can you make the BPF and notch outputs verbose as well?
138. *(Shared BPF results for rag_pipeline.)* — *(Reviewing BPF output.)*
-

Phase 11 — Notch Template Injection, Resume & Final Runs

139. notch --sample-per-topology 6 --verbose — *(Shared traceback showing NoneType format error.)*
140. *(Shared verbose run output with timing.)* — Look at the above outputs and tell me how much time would it take to run a test for 1000 points.
141. cd Course_Projects/CPSC5770 *(Shared existing job script.)* — Modify this to run a test for 1000 points on the rag_final_pipeline.
142. I ran some tests again but it didn't finish, can you modify the script so that when I run the BPF and Notch again, it will continue where I left off?
143. I ran some more tests, can you see the results?
144. I re-ran the baseline I think, can you check if I re-ran it?
145. No no, I meant the new baseline with the accurate metric checking.
146. Check the output folder in New_Datagen.
-

Phase 12 — Report Figures

147. Okay let's make some graphs for the final report with these results!
148. Yes please. *(Re: adding baseline comparison chart.)*
149. *(Shared image of broken Fig 5 — 1631×54179 pixel strip.)* — *(Requesting fix.)*
-

Phase 13 — Interactive Chat Interface & NL2SPICE Repo

150. Can you make a chat-type interface for the final pipeline and also plot the frequency response of the final generated netlist?
151. I don't think we should ask the user what topology they want, it needs to be inferred from the frequencies and the attenuation right?

152. I have cloned this repo here /home/scm93/project_pi_asn52/scm93/NL2SPICE. Can we copy paste the relevant files over to the repo and add a README file that explains everything? Also feel free to reorganize the file structure if you want to.

Phase 14 — Documentation

153. Okay now can you give me a list of prompts I have asked you in this entire conversation? I need it for my report. You can add it as a README file in CPSC5770.

Total: 153 prompts across 14 development phases.